

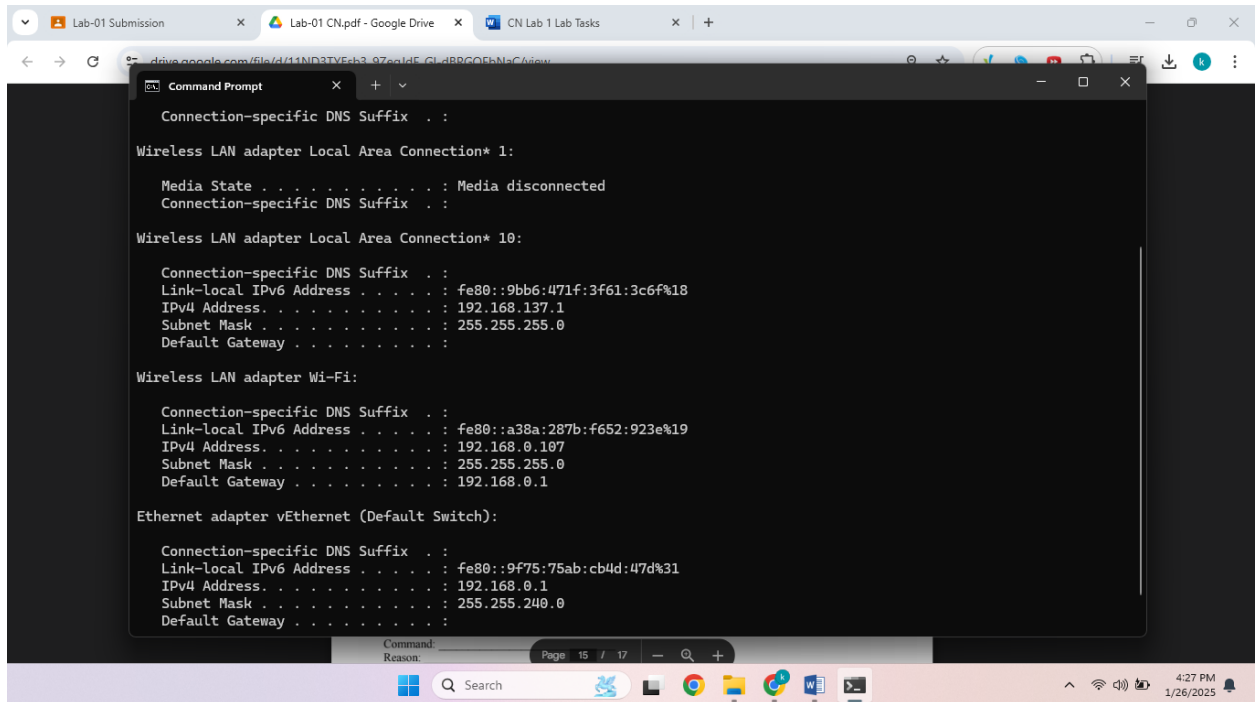
CN Lab 1 Lab Tasks

22k-5195 Laiba Fatima

1. Find the IP address of the computer you are currently using.

Command: **ipconfig**

IP Address: **192.168.0.107**



```
Connection-specific DNS Suffix . :  
  
Wireless LAN adapter Local Area Connection* 1:  
  
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :  
  
Wireless LAN adapter Local Area Connection* 10:  
  
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::9bb6:471f:3f61:3c6f%18  
IPv4 Address. . . . . : 192.168.137.1  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :  
  
Wireless LAN adapter Wi-Fi:  
  
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::a38a:287b:f652:923e%19  
IPv4 Address. . . . . : 192.168.0.107  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 192.168.0.1  
  
Ethernet adapter vEthernet (Default Switch):  
  
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::9f75:75ab:cb4d:47d%31  
IPv4 Address. . . . . : 192.168.0.1  
Subnet Mask . . . . . : 255.255.240.0  
Default Gateway . . . . . :
```

2. Find the IP address of the computer you are currently using, plus MAC address, the gateway, plus whether DHCP is turned on.

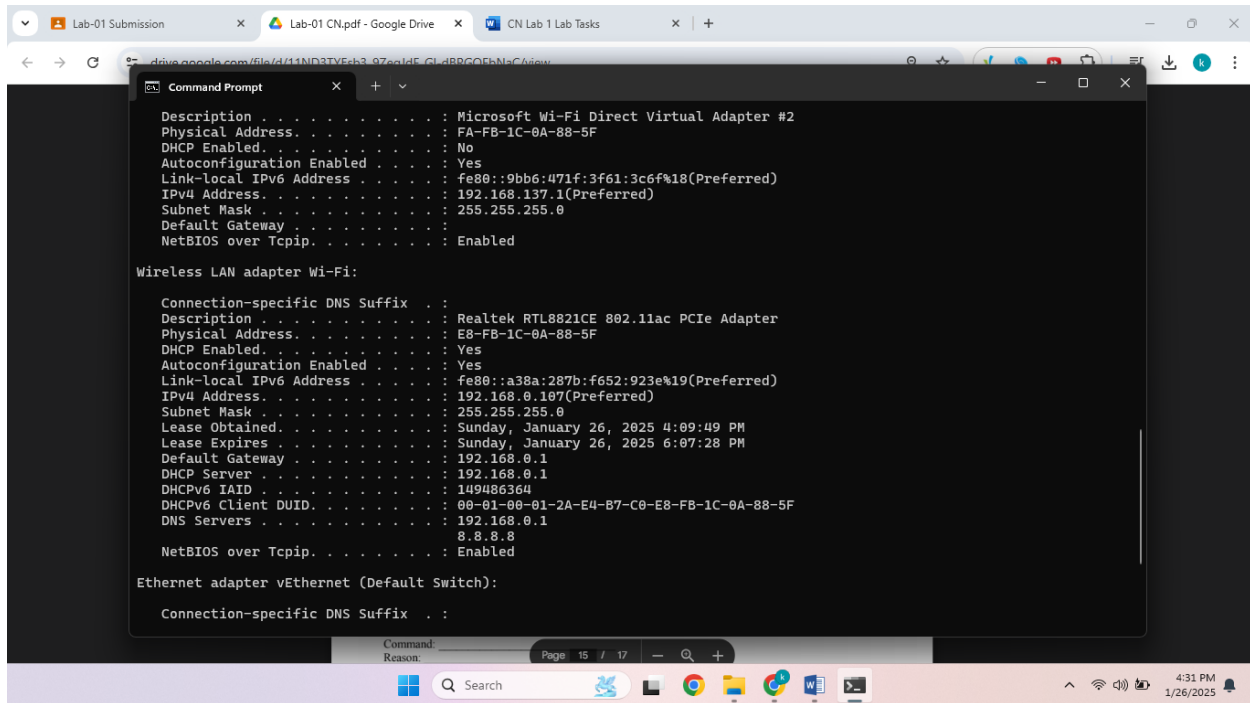
Command: **ipconfig**

IP Address: **192.168.0.107**

MAC Address: **E8-FB-1C-0A-88-5F**

Gateway: **192.168.0.1**

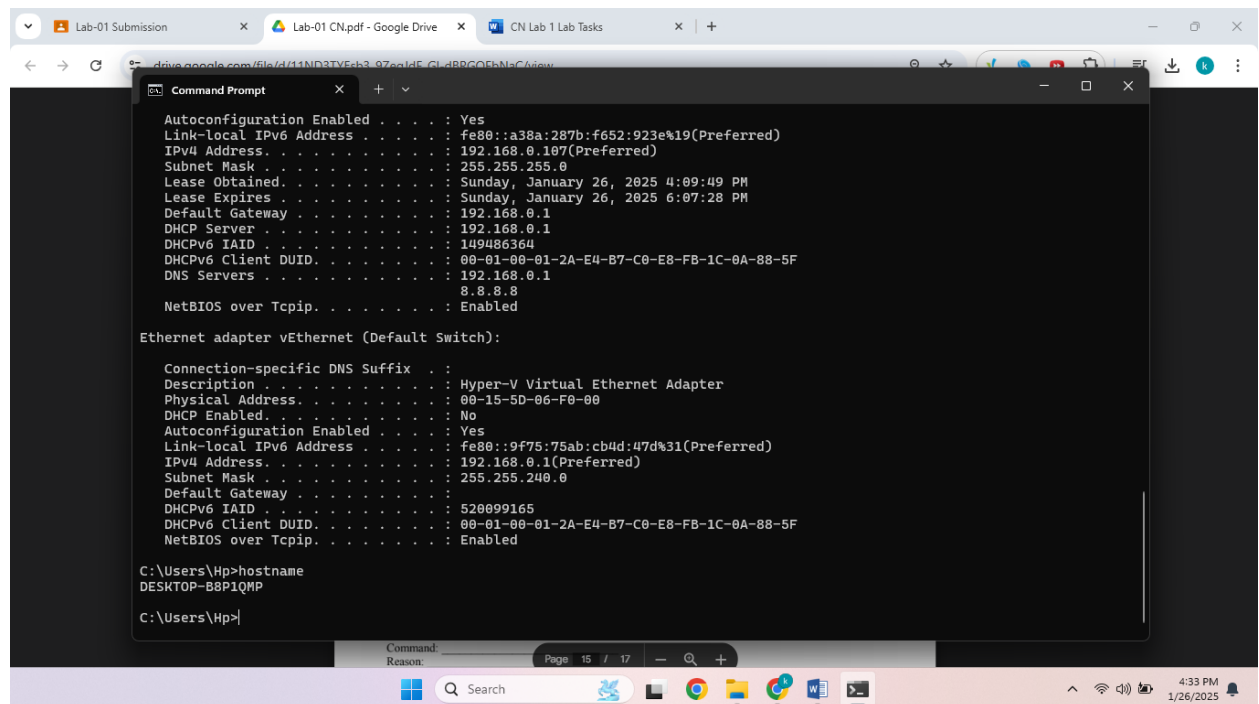
DHCP: **Enabled**



3. Display the host name of the computer.

Command: **hostname**

Name: **DESKTOP-B8P1QMP**



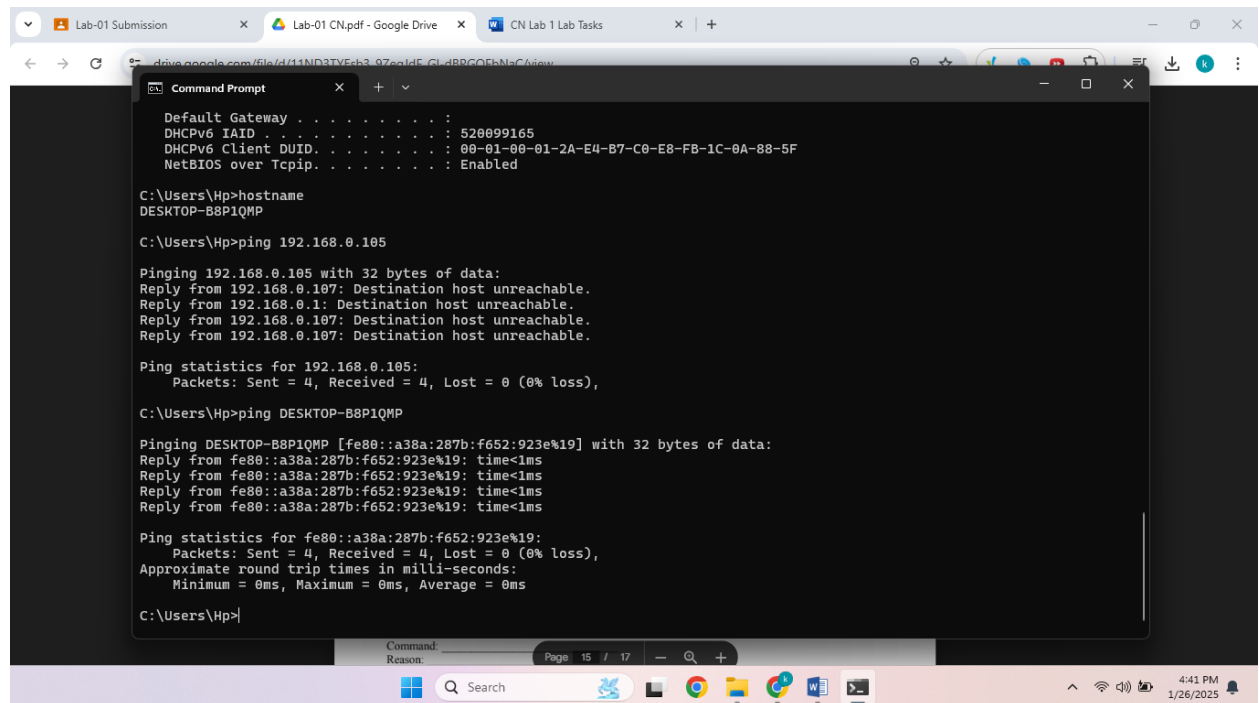
4. Check for basic IP connectivity between two computers by name and IP address.
How can basic IP connectivity be checked? What are the reasons why there is no connectivity?

Command: **ping 192.168.0.105 | ping DESKTOP-B8P1QMP**

Reason: **1. Incorrect address/name**

2. wireless network not connected/ network cable unplugged

3. Requests are blocked by firewall



The screenshot shows a Windows Command Prompt window with the following output:

```
Default Gateway . . . . . :  
DHCPv6 IAID . . . . . : 520099165  
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-E4-B7-C0-E8-FB-1C-0A-88-5F  
NetBIOS over Tcpip. . . . . : Enabled  
  
C:\Users\Hp>hostname  
DESKTOP-B8P1QMP  
  
C:\Users\Hp>ping 192.168.0.105  
  
Pinging 192.168.0.105 with 32 bytes of data:  
Reply from 192.168.0.107: Destination host unreachable.  
Reply from 192.168.0.1: Destination host unreachable.  
Reply from 192.168.0.107: Destination host unreachable.  
Reply from 192.168.0.107: Destination host unreachable.  
  
Ping statistics for 192.168.0.105:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
  
C:\Users\Hp>ping DESKTOP-B8P1QMP  
  
Pinging DESKTOP-B8P1QMP [fe80::a38a:287b:f652:923e%19] with 32 bytes of data:  
Reply from fe80::a38a:287b:f652:923e%19: time<1ms  
Reply from fe80::a38a:287b:f652:923e%19: time<1ms  
Reply from fe80::a38a:287b:f652:923e%19: time<1ms  
Reply from fe80::a38a:287b:f652:923e%19: time<1ms  
  
Ping statistics for fe80::a38a:287b:f652:923e%19:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 0ms, Average = 0ms  
  
C:\Users\Hp>
```

5. Find out which ports on your host are connected to applications. Connect the browser to some external web page before running the appropriate command.

Command: **netstat -ano**

```
C:\Users\Hp>start https://www.google.com

C:\Users\Hp>netstat -ano

Active Connections

Proto Local Address           Foreign Address         State       PID
TCP 0.0.0.0:135             0.0.0.0:0               LISTENING   1572
TCP 0.0.0.0:445             0.0.0.0:0               LISTENING   4
TCP 0.0.0.0:2179           0.0.0.0:0               LISTENING   2584
TCP 0.0.0.0:2869           0.0.0.0:0               LISTENING   4
TCP 0.0.0.0:5040           0.0.0.0:0               LISTENING   8420
TCP 0.0.0.0:5357           0.0.0.0:0               LISTENING   4
TCP 0.0.0.0:7250           0.0.0.0:0               LISTENING   11160
TCP 0.0.0.0:7680           0.0.0.0:0               LISTENING   16512
TCP 0.0.0.0:49664          0.0.0.0:0               LISTENING   1264
TCP 0.0.0.0:49665          0.0.0.0:0               LISTENING   1080
TCP 0.0.0.0:49668          0.0.0.0:0               LISTENING   2340
TCP 0.0.0.0:49669          0.0.0.0:0               LISTENING   3312
TCP 0.0.0.0:49672          0.0.0.0:0               LISTENING   4920
TCP 0.0.0.0:49676          0.0.0.0:0               LISTENING   1192
TCP 127.0.0.1:5354         0.0.0.0:0               LISTENING   1768
TCP 127.0.0.1:5354         127.0.0.1:50162         ESTABLISHED 1768
TCP 127.0.0.1:5354         127.0.0.1:50163         ESTABLISHED 1768
TCP 127.0.0.1:27015       0.0.0.0:0               LISTENING   12364
TCP 127.0.0.1:49677       127.0.0.1:49678         ESTABLISHED 1480
TCP 127.0.0.1:49678       127.0.0.1:49677         ESTABLISHED 1480
TCP 127.0.0.1:49679       127.0.0.1:49680         ESTABLISHED 1688
TCP 127.0.0.1:49680       127.0.0.1:49679         ESTABLISHED 1688
TCP 127.0.0.1:49681       0.0.0.0:0               LISTENING   4592
TCP 127.0.0.1:49681       127.0.0.1:51101         ESTABLISHED 4592
TCP 127.0.0.1:49681       127.0.0.1:51140         ESTABLISHED 4592
```

6. Find the path of routers to www.google.com. What is its IP address? How many hops involved in the path?

Command: **tracert www.google.com**

30 hops

```
UDP [::]:61856             *: *                2348
UDP [::]:63233             *: *                3996
UDP [::1]:1900             *: *                6904
UDP [::1]:5353             *: *                1768
UDP [::1]:52560            *: *                6904
UDP [fe80::9bb6:471f:3f61:3c6f%18]:1900 *: *                6904
UDP [fe80::9bb6:471f:3f61:3c6f%18]:52559 *: *                6904
UDP [fe80::9f75:75ab:cb4d:47d%31]:1900 *: *                6904
UDP [fe80::9f75:75ab:cb4d:47d%31]:52562 *: *                6904
UDP [fe80::a38a:287b:f652:923e%19]:1900 *: *                6904
UDP [fe80::a38a:287b:f652:923e%19]:52561 *: *                6904

C:\Users\Hp>tracert www.google.com

Tracing route to www.google.com [172.217.19.228]
over a maximum of 30 hops:

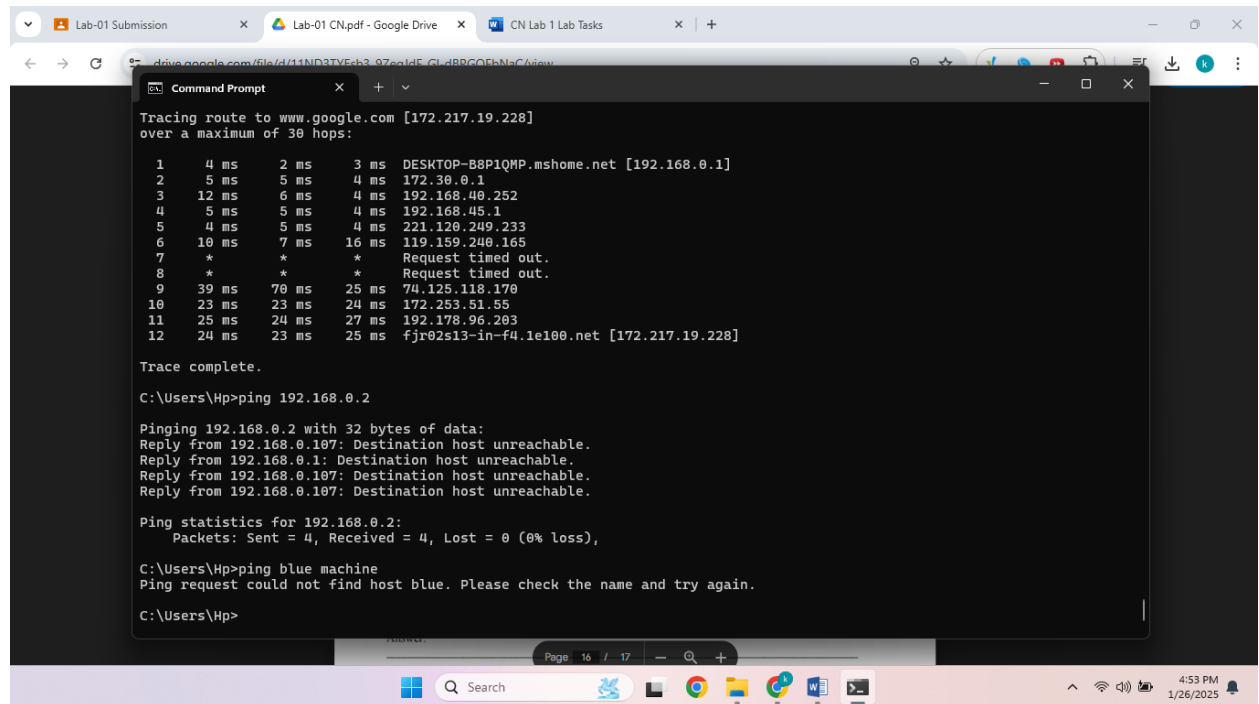
 1  4 ms  2 ms  3 ms  DESKTOP-B8P1QMP.mshome.net [192.168.0.1]
 2  5 ms  5 ms  4 ms  172.30.0.1
 3  12 ms  6 ms  4 ms  192.168.40.252
 4  5 ms  5 ms  4 ms  192.168.45.1
 5  4 ms  5 ms  4 ms  221.120.249.233
 6  10 ms  7 ms  16 ms  119.159.240.165
 7  *  *  *  Request timed out.
 8  *  *  *  Request timed out.
 9  39 ms  70 ms  25 ms  74.125.118.170
10  23 ms  23 ms  24 ms  172.253.51.55
11  25 ms  24 ms  27 ms  192.178.96.203
12  24 ms  23 ms  25 ms  fjr02s13-in-f4.1e100.net [172.217.19.228]

Trace complete.

C:\Users\Hp>
```

7. A ping to 192.168.0.2 works but a ping to the machine's name "blue machine" fails. What could be wrong?

Reason: **Machine's name is not registered with the DNS server**



```
Tracing route to www.google.com [172.217.19.228]
over a maximum of 30 hops:
  0  4 ms  2 ms  3 ms  DESKTOP-B8P1QMP.mshome.net [192.168.0.1]
  1  5 ms  5 ms  4 ms  172.30.0.1
  2  12 ms  6 ms  4 ms  192.168.40.252
  3  5 ms  5 ms  4 ms  192.168.45.1
  4  4 ms  5 ms  4 ms  221.120.249.233
  5  10 ms  7 ms  16 ms  119.159.240.165
  6  *      *      *      Request timed out.
  7  *      *      *      Request timed out.
  8  39 ms  70 ms  25 ms  74.125.118.170
  9  23 ms  23 ms  24 ms  172.253.51.55
 10  25 ms  24 ms  27 ms  192.178.96.203
 11  24 ms  23 ms  25 ms  fjr02s13-in-f4.1e100.net [172.217.19.228]
 12

Trace complete.

C:\Users\Hp>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:
Reply from 192.168.0.107: Destination host unreachable.
Reply from 192.168.0.1: Destination host unreachable.
Reply from 192.168.0.107: Destination host unreachable.
Reply from 192.168.0.107: Destination host unreachable.

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\Hp>ping blue machine
Ping request could not find host blue. Please check the name and try again.

C:\Users\Hp>
```

8. Which type of cable will you use to connect in a normal home installation?

Answer: **Cat5e or Cat6 (Twisted Pair Cable)**

9. Can you connect a Switch to another Switch or a router to a PC using a straight-through cable? Explain your answer.

Answer: **Straight cable: for different types of devices so router to pc is fine.**

Crossover Cable: for same type of devices so Switch to switch is possible using this cable.

10. Write a brief report on your home network or any organizational network including topology, 1 page max).

Answer: **My network follows a star topology, where all devices (laptops, mobiles, tablet, printer) are connected to a router using Wi-Fi. Network is protected by WPA2 encryption to secure wireless communications.**

11. What type of cable would you use to connect: a) A PC to a switch. b) A router to another router. Explain why the chosen cable is suitable for each scenario.

Answer: **a) Straight cable: for different types of devices so pc to switch is okay.**

b) Crossover Cable: for same type of devices so Router to Router is possible.

12. Describe the difference between half-duplex and full-duplex communication.

Provide an example of a device or scenario where each mode is used.

Answer: **Half-Duplex: Data can travel in both directions, but not at the same time, e.g. Walkie-talkies.**

Full-Duplex: Data can travel in both directions simultaneously, allowing two-way communication, e.g. Telephones

13. Explain the purpose of the following network devices and give a real-world example of their usage: a) Hub b) Switch c) Router

Answer: **a) A hub is a networking device that broadcasts data to all connected devices within a network, e.g. Used in office networks to connect multiple devices to a single main server computer.**

b) A switch directs data to the specific device it is intended for, based on MAC addresses, e.g. Used in modern office networks to connect computers, printers, and servers, ensuring data goes only to the intended device.

c) A router connects different networks and routes data between them based on IP addresses, e.g. home routers connect devices to the internet

14. What is the function of a MAC address? How does it differ from an IP address? Use an example to illustrate their roles in a network.

Answer: **MAC Address: Unique to the device, used for communication within a local network. E.g. a computer sends data packets to the MAC address of a printer to ensure the packet reaches the correct device.**

IP Address: Used for routing between different networks, typically assigned. E.g. computer uses website's IP address to send a request to the web server's IP address.

15. Design a simple network for a small office with 10 computers, 2 printers, and a single internet connection. Indicate the devices and connections needed, and explain why each component is essential.

Answer: **We need a router to connect the network to the internet and provide security. A switch connects the computers and printers within the office to communicate. A Wi-Fi access point can be added for wireless connectivity. The computers and printers connected via Ethernet to the switch.**

